CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

First Semester of M. Tech. (ME) Examination

December 2012

ME702.01 Advances in Manufacturing Technology

Date: 20.12.2012, Thursday **Time: 10:00 a.m. To 01:00 p.m.** **Maximum Marks: 70**

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION – I

Q - 1 (a) Answer following multiple choice type questions. There could be more than one correct answer. Write all correct choices in answer sheet. **[10]**

(i) Name welding process(es) that do not require filler material:
(p) Metal inert gas welding (q) Resistance welding
(r) Soldering (s) Electroslag welding

(ii) In resistance welding the resistance should be highest:
(p) At interface between workpiece and electrode (q) Inside electrode
(r) At interface between two workpieces (s) Inside workpiece

(iii) Besides protecting the weld from oxidation, shielding also prevents
(p) Arc blow (q) Weld inclusions
(r) Weld splatter (s) All of the above

(iv) Which of the followings is(are) solid state welding processes
(p) Resistance welding (q) Gas welding
(r) Ultrasonic welding (s) Pressure Welding

(v) A fusion welding process using chemical energy source (other than gas welding) is:
(p) Arc welding (q) Induction welding
(r) Thermit welding (s) None of the above

(vi) Which of the following process use a dispensable mold as well as dispensable pattern:
(p) Sand casting (q) Centrifugal casting
(r) Shell molding (s) Investment Casting

(vii) Solid metal contraction is compensated by providing:
(p) Additional feeder (q) Additional gates
(r) Additional sprue (s) Shrinkage allowances in pattern

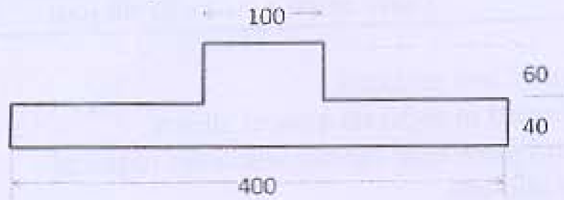
(viii) Stresses on the mold due to core weight and buoyancy force will depends on:
(p) Core surface area (q) Core print surface area
(r) Core and core print surface area (s) Core print projected surface area

(ix) Insulated feeder have feeding capacity in the range of:
(p) 10 to 15 percentage (q) 15 to 25 percentage
(r) 25 to 40 percentage (s) 30 to 50 percentage

(x) In general, which of the following is ideal condition for effective feeding (M is modulus):
(p)$M_{feeder} > M_{neck} > M_{hot\ spot}$ (q) $M_{hot\ spot} > M_{neck} > M_{feeder}$
(r) $M_{feeder} \geq M_{neck} \geq M_{hot\ spot}$ (s) $M_{hot\ spot} \geq M_{neck} \geq M_{feeder}$

Q - 1 (b) List various methods of micromachining and explain any one in detail. **[05]**

Q - 2 A solid steel wheel (middle cross-section shown below) is to be produced by sand casting process. **[08]**
(i) Design a cylindrical feeder connected to top of middle hub, without any neck, if modulus ratio = 1.25, and feeder H/D = 2. Account for increased modulus of casting after feeder connection.
(ii)What is the yield for this casting?



OR

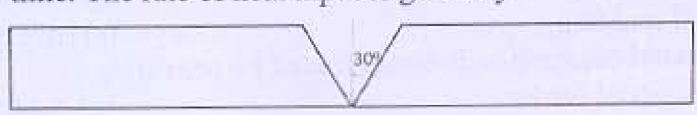
Q - 2 The voltage-length characteristic of a direct current arc is given by, $V= 20+40l$ *volts*, where l is length of arc in *cm*. Assuming a linear power source characteristic with an open circuit voltage *80 V* and a short circuit current *1000 amp.*, determine the optimal arc power and corresponding arc power. **[08]**

Q - 3 (a) Welding time to join a flange to pressure vessel by different operators and welding processes is given in following table. Set up the ANOVA table and analyze the results by commenting on significance of individual parameter and their interactions. F*value* required to achieve given confidence level are also provided in table. **[06]**

Operator	Welding Process		Confidence Level	F*value* (1,4)
	P1	P2	90%	4.45
O1	6,8	3,4	95%	7.71
O2	7,8	9,10	99%	21.2

Q - 3 (b) Find the maximum possible welding speed during butt welding of two steel plates of *6 mm* thickness and *80 mm* long (shown below) by arc welding, if arc power is *10 kVA*. Assume arc is on for 85% of time. The rate of heat input is given by **[06]**

$$Q = 1.2kT_m t\left(8.0+\frac{vw}{\alpha}\right)$$



where, t = plate thickness, w = width of weld, v = welding speed, and α = thermal diffusivity ($K/\rho C_p$). Take $k = 80$ *W/m-K*, $\rho = 8000$ *kg/m*^{3}, $C_p = 400$ *J/kg-K*, $T_m = 1530°C$, and ambient temperature = $30°C$.

OR

Q - 3 (a) In a three factor two level experimental data as shown below, estimate individual and interaction effects. Develop mathematical model (regression equation) of the problem. **[06]**

Order	Factor A	Factor B	Factor C	Response
1	-1	-1	-1	12
2	1	-1	-1	10
3	-1	1	-1	2
4	1	1	-1	5
5	-1	-1	1	3
6	1	-1	1	7
7	-1	1	1	3
8	1	1	1	8